

Emergent Magnetism in Finite Nanographenes

Topological end states in 7-AGNRs and a designed Heisenberg spin chain in triangulene–phenalenyl copolymers

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7-Atom-Wide Armchair Graphene Nanoribbons

Triangulene–Phenalenyl Nanographene Copolymer

Outlook

7-Atom-Wide Armchair Graphene Nanoribbons

Motivation: topological end states in finite 7-AGNRs

7-AGNRs belong to the $N = 3p + 1$ family ($p = 2$): semiconducting, synthesized by on-surface polymerization/cyclodehydrogenation of DBBA precursors on Au(111).

Finite-length ribbons host **spin-polarized topological end states** (SPTs) at their zigzag-terminated ends, each carrying $S \approx 1/2$.

Origin: a nontrivial $\mathbb{Z}_2$ (Zak-phase) invariant of the bulk bands $\Rightarrow$ bulk–boundary correspondence guarantees in-gap boundary modes, independent of edge details.

Key question: what is the exchange coupling $J(L)$ between the two end spins, and how does it scale with ribbon length L (number of DBBA monomers)?

Third-nearest-neighbour tight-binding on the p_z lattice:

$$H_{\text{TB}} = - \sum_{\langle i,j \rangle} t_1 c_i^\dagger c_j - \sum_{\langle\langle i,j \rangle\rangle} t_2 c_i^\dagger c_j - \sum_{\langle\langle\langle i,j \rangle\rangle\rangle} t_3 c_i^\dagger c_j + \text{H.c.}$$

t_1	t_2	t_3	U (Hubbard)
-2.7 eV	-0.2 eV	-0.18 eV	3.0 eV

Mean-Field Hubbard (MFH): $H = H_{\text{TB}} + U \sum_i n_{i\uparrow} n_{i\downarrow}$, decoupled self-consistently (SCF tolerance 10^{-13} on the density matrix).

$$m_i = \langle n_{i\uparrow} \rangle - \langle n_{i\downarrow} \rangle \quad (\text{local magnetic moment})$$

Ground-state selection: Lieb's theorem

Lieb's theorem (1989)

For the repulsive Hubbard model at half-filling on a bipartite lattice, the ground-state total spin is

$$S = \frac{1}{2}|N_A - N_B|,$$

where N_A, N_B are the sublattice populations.

A pristine 7-AGNR is a *balanced* bipartite lattice, $N_A = N_B$.

⇒ The ground state is guaranteed to be a total **singlet** ($S = 0$): the AFM open-shell configuration, not the FM triplet.

This rigorous result is used below to identify which of the two computed states (AFM/FM) is physical.

Two independent routes to the exchange coupling J

1. Self-consistent MFH total energies

$$J = E_{\text{FM}} - E_{\text{AFM}}$$

AFM: open-shell singlet, $S_z = 0$. FM: triplet enforced via $q = (N/2+1, N/2-1)$, $S_z = 1$.

2. Effective Hubbard dimer (Anderson superexchange)

$$t_{\text{eff}} = \frac{1}{2}|\varepsilon_{\text{LUMO}} - \varepsilon_{\text{HOMO}}|, \quad U_{\text{eff}} = U \sum_i |\psi_L(i)|^4, \quad J = \frac{4 t_{\text{eff}}^2}{U_{\text{eff}}}$$

$\psi_L = (\psi_{\text{HOMO}} + \psi_{\text{LUMO}})/\sqrt{2}$: reconstructed localized end-state orbital.

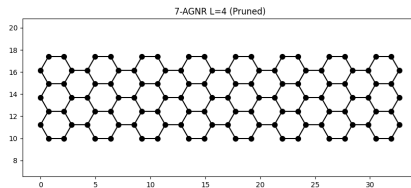
Since $t_{\text{eff}} \propto e^{-L/\xi_0}$: $J(L) = Ae^{-L/\xi}$ is predicted to decay exponentially.

Geometry construction and definition of L

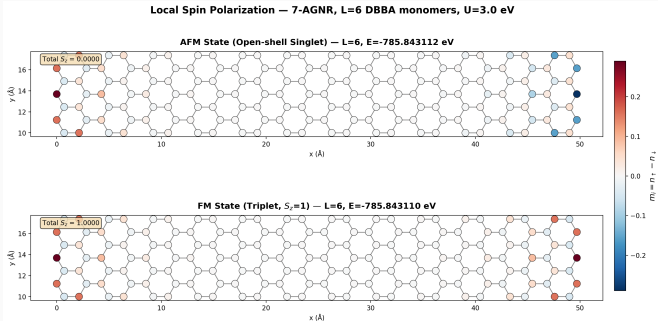
Tile the periodic 7-AGNR unit cell,
remove periodic boundary
conditions.

Iteratively prune coordination-1
(dangling-bond) atoms $\Rightarrow$ clean
zigzag termination.

Each DBBA monomer contributes
28 C atoms; L = number of
monomers incorporated.



Spin-polarized end states: AFM vs. FM ($L = 6$)



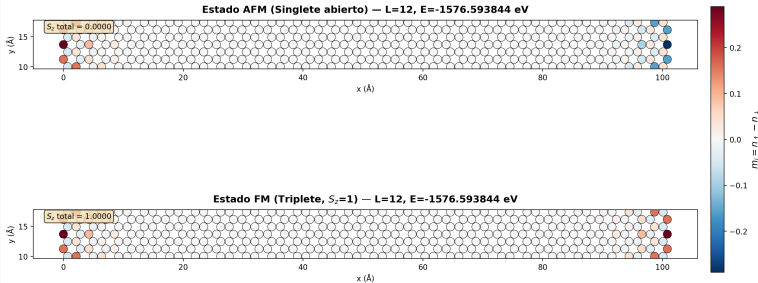
Spin density strongly localized at the two zigzag ends; bulk atoms unpolarized.

AFM (top): opposite sign at the two ends, $S_z = 0$ – open-shell singlet. FM (bottom): same sign, $S_z = 1$.

At $L = 6$: $J = E_{\text{FM}} - E_{\text{AFM}} \approx 2 \mu\text{eV}$.

Robustness at larger length ($L = 12$)

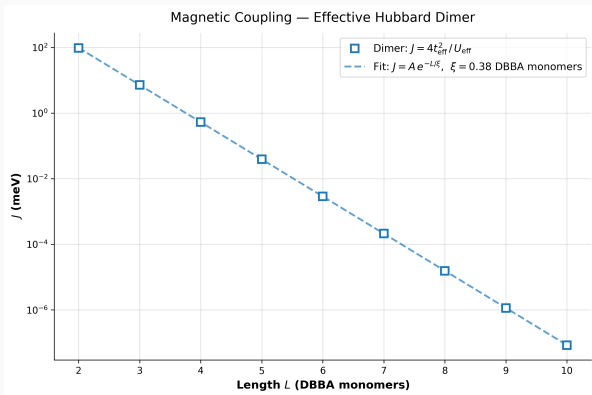
Polarización de Espín Local — 7-AGNR, $L=12$ antracenos, $U=3.0$ eV



End-state localization persists unchanged as L grows – confirms the topological (not finite-size-artifact) origin of the SPTs.

The AFM/FM splitting shrinks further, foreshadowing the numerical challenge of resolving J for long ribbons.

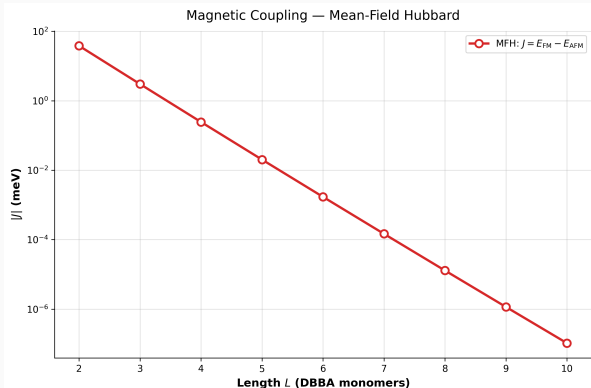
Method 1: exchange coupling from the effective dimer



Perfectly clean exponential decay:
 $J(L) = A e^{-L/\xi}$, fitted $\xi = 0.38$
DBBA monomers.

Analytic diagonalization of the
non-interacting Hamiltonian: no
self-consistency, robust to
arbitrarily small J .

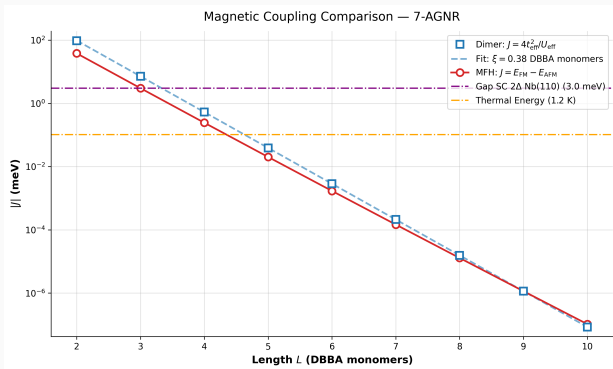
Method 2: exchange coupling from self-consistent MFH



$J = E_{\text{FM}} - E_{\text{AFM}}$ from converged total energies: the direct many-body observable.

Tracks the same exponential decay while the splitting exceeds the SCF numerical precision.

Comparison: quantitative agreement between methods

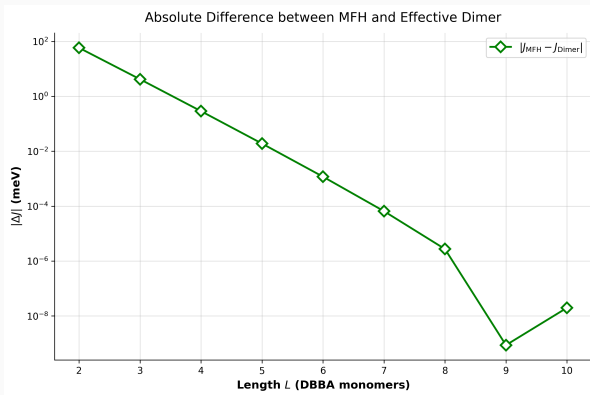


Dimer (superexchange) and MFH (total-energy) methods agree quantitatively over the overlapping length range.

Reference scales shown: SC gap $2\Delta \approx 3.0$ meV of Nb(110); thermal energy of an STM at $T = 1.2$ K.

Both set practical bounds on experimentally resolvable J .

Deviation between methods and the MFH noise floor



$|\Delta J| = |J_{\text{MFH}} - J_{\text{Dimer}}|$ itself decays exponentially while both methods remain consistent.

For long ribbons, J_{MFH} drops below the SCF precision ($\sim 10^{-13}$ on the density $\Rightarrow \sim 10^{-10}$ meV) and saturates at a noise floor.

The dimer model is unaffected and is preferred for long ribbons.

Finite 7-AGNRs host two topologically protected $S = 1/2$ end states ($\mathbb{Z}_2 = 1$); the AFM open-shell singlet is the ground state (Lieb's theorem).

The intramolecular exchange coupling decays exponentially, $J(L) = Ae^{-L/\xi}$, with $\xi \approx 0.38$ monomer units – tunable over many orders of magnitude via L .

The effective Hubbard dimer (Anderson superexchange) faithfully reproduces the self-consistent MFH results and extends beyond its numerical precision limit.

Establishes a validated, computationally efficient route to design length-tunable magnetic couplings in GNR-based molecular spintronics.

Triangulene–Phenalenyl Nanographene Copolymer

Motivation: chemically designed spin chains

Complementary strategy to ribbon end states: assemble a spin chain from **chemically localized radical building blocks**.

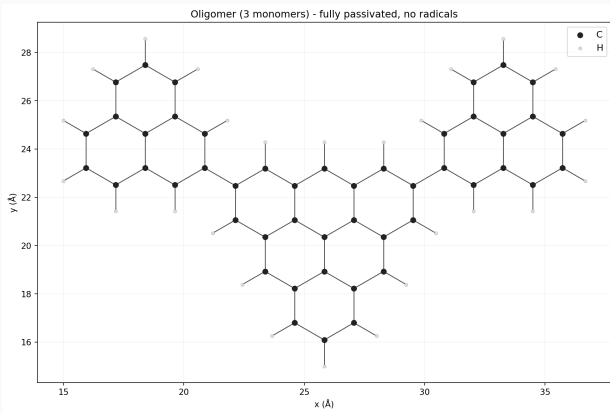
Repeating motif $P-T-P$ under open boundary conditions: $P-T-P-[P-T-P]_{N-1}$.

Phenalenyl (P): alternant radical, sublattice imbalance 1 $\Rightarrow$ single $S = \frac{1}{2}$ moment.

[3]Triangulene (T): sublattice imbalance 2 $\Rightarrow S = 1$ moment; represented as two $S = \frac{1}{2}$ sites T_a, T_b locked into a triplet by a strong *ferromagnetic* intra-unit exchange $J_T < 0$.

Every monomer maps onto four $S = \frac{1}{2}$ nodes $[P, T_a, T_b, P]$; the oligomer maps onto a chain of $n = 4N$ spin- $\frac{1}{2}$ sites.

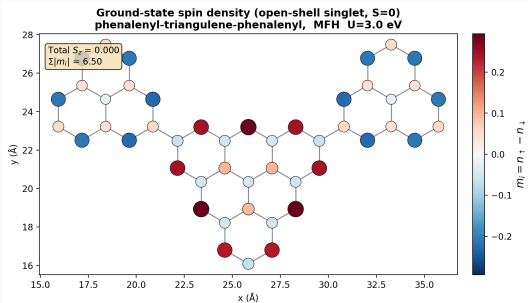
Geometry: closed-shell reference oligomer



Fully passivated ($N = 3$) oligomer used to validate the geometric construction before introducing the open-shell radical character.

The corresponding `.xyz` geometry is the direct input to the MFH spin-density calculation.

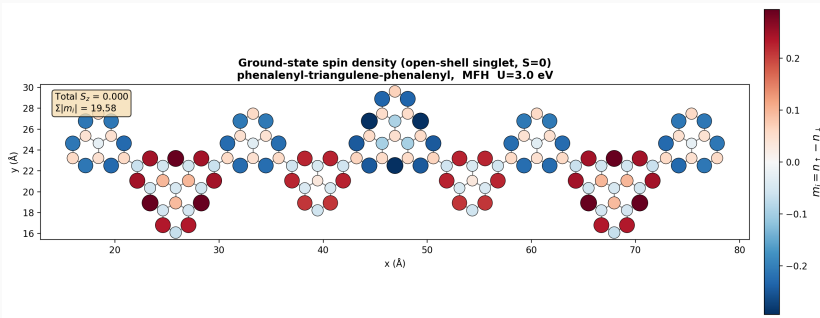
MFH ground-state spin density: single monomer ($N = 1$)



Open-shell singlet ground state,
 $S_z = 0$ (MFH, $U = 3.0$ eV): total
 $\sum_i |m_i| = 6.50$.

Spin density directly visualizes the P
($S=1/2$) and T ($S=1$, on T_a, T_b)
building blocks predicted by
sublattice-imbalance counting.

MFH ground-state spin density: trimer ($N = 3$)

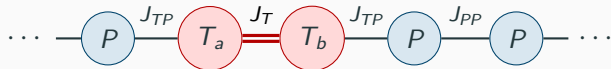


Same open-shell singlet character ($S_z = 0$), now $\sum_i |m_i| = 19.58$ – radical character grows with chain length.

Alternating sign along the chain: the atomistic fingerprint of the antiferromagnetic P – T and P – P couplings encoded below.

From MFH to an effective Heisenberg spin chain

$$H = \sum_{m=0}^{N-1} \left[J_{TP} (\mathbf{S}_{4m} \cdot \mathbf{S}_{4m+1} + \mathbf{S}_{4m+2} \cdot \mathbf{S}_{4m+3}) + J_T \mathbf{S}_{4m+1} \cdot \mathbf{S}_{4m+2} \right] + J_{PP} \sum_{m=0}^{N-2} \mathbf{S}_{4m+3} \cdot \mathbf{S}_{4m+4}$$



J_{PP} (AFM)	J_{TP} (AFM)	J_T (FM)
38 meV	40 meV	-500 meV

Since $|J_T| \gg J_{TP}, J_{PP}$: each (T_a, T_b) pair freezes into an effective $S = 1$ object $\Rightarrow$
 mixed $\frac{1}{2}-1-\frac{1}{2}$ AFM spin chain.

Exact diagonalization: method and scaling

H is $SU(2)$ -symmetric and one-dimensional $\Rightarrow$ ED gives the numerically exact ground state, spectrum, and total spin S, S_z with no mean-field bias.

Implemented in QuTiP: each spin operator built as a tensor product of Pauli matrices, H assembled bond by bond.

Hilbert space $\dim \mathcal{H} = 2^N$: 4096 states at $N = 3$, 65536 at $N = 4$.

Dense ED feasible to $n \sim 14$ – 16 sites; conserving total S_z + sparse Lanczos/Arnoldi extends this to $n \sim 18$ – 20 .

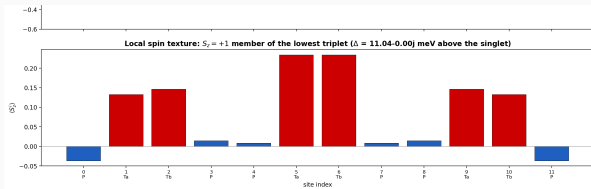
Lieb–Mattis theorem

A balanced bipartite chain ($S_A = S_B$) has a total-singlet ($S = 0$) ground state for every N .

$\langle S_z^i \rangle$ vanishes identically in the ground state by symmetry: no local magnetization is visible there.

The intrinsic magnetic texture is instead revealed by the $S_z = +1$ member of the lowest $S = 1$ multiplet – the observable used on the next slide.

ED results: ground state and spin texture ($N = 3$)



Ground state: total singlet, $S = 0$,
 $E_0 = -514.43$ meV.

Lowest triplet ($S_z = +1$): large
moments on T_a , T_b
(~ 0.13 – 0.23), small on P ,
alternating sign between
successive triangulene units.

Singlet–triplet gap
 $\Delta = 11.04$ meV.

The P – T – P copolymer is a chemically programmed realization of a mixed $\frac{1}{2}$ – 1 – $\frac{1}{2}$ antiferromagnetic Heisenberg chain, with couplings inherited from atomistic MFH/second-order exchange.

MFH spin-density maps and exact-diagonalization spin textures agree qualitatively: the same alternating pattern, from the fully atomistic to the effective spin model.

$S = 0$ ground state guaranteed by the Lieb–Mattis theorem; finite singlet–triplet gap (11.04 meV at $N = 3$) set by J_{PP} and J_{TP} .

Linear scaling $n = 4N$ of the spin lattice with oligomer length makes the model straightforwardly extensible via sparse ED.

Outlook

Two complementary routes to nanographene magnetism

7-AGNR end states

Magnetism from topology: $\mathbb{Z}_2$ -protected zero modes at ribbon termini.

Coupling set by ribbon *length* L ;
exponential tunability, $\xi \approx 0.38$
monomers.

Validated by Hubbard-dimer
superexchange beyond MFH's numerical
reach.

Triangulene–phenalenyl chain

Magnetism from chemistry:
sublattice-imbalance radicals (P, T) as
designed spin- $\frac{1}{2}$ /spin-1 units.

Couplings set by *precursor design*;
scalable $4N$ -site Heisenberg chain.

Validated by exact diagonalization
against MFH spin densities.

Outlook

Both frameworks combine atomistic MFH with an exactly solvable low-energy model – a template for engineering length- and chemistry-tunable magnetic couplings for molecular spintronics and quantum information.

Thank you

Questions & discussion